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(MCI-186) in amyotrophic lateral sclerosis patients. *Amyotroph Lateral Scler Frontotemporal Degener*. 2014;15(7–8):610-617.
17. Servais L, Deconinck N, Moraux A, et al. Innovative methods to assess upper limb strength and function in non-ambulant Duchenne patients. *Neuromuscul Disord*. 2013;23(2):139-148.
18. Wachter NJ, Mentzel M, Hutz R, Krischak GD, Gulke J. Quantification of weakness caused by distal median nerve lesion by dynamometry. *J Hand Ther*. 2018;31(1):74-79.
19. Li K, Wei N, Cheng M, Hou X, Song J. Dynamical coordination of hand intrinsic muscles for precision grip in diabetes mellitus. *Sci Rep*. 2018;8(1):4365.

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Single-institutional reference values for concentric needle jitter analysis using the extrapolated reference values procedure: Comparison to published reference values

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Abstract

Background: The extrapolated reference values procedure (E-Ref) was used to compare data from a single institution with the recently published reference value (RV) for concentric electrode jitter.

Methods: Data from voluntarily activated concentric needle jitter studies in the frontalis muscle were obtained using retrospective chart review. All measured signals were reviewed for acceptable quality. Cutoff values for increased jitter were calculated using E-Ref, and compared with the published RVs.

Results: At total of 1501 apparent single-fiber action potential (ASFAP) pairs were reviewed; 1371 ASFAP pairs were determined to have acceptable quality. The cutoff value identified by E-Ref from all reviewed ASFAP pairs was 36 microseconds and the cutoff for acceptable pairs was 35 microseconds. Using either of these cutoff values (36 or 35 microseconds) did not result in a significant difference in percentage of jitter recordings considered normal when compared with the recently published RV (38 microseconds).

Discussion: The single-institution jitter cutoff value obtained by E-Ref gives results that are not significantly different from the reported RV.

KEYWORDS

concentric needle, EMG, jitter, myasthenia gravis, reference values, SFEMG

Abbreviations: AP, action potential; ASFAP, apparent single-fiber action potential; CNE, concentric needle electrode; E-Ref, extrapolated reference values procedure; IQR, interquartile range; MCD, mean consecutive difference; RV, reported reference value; SFE, single-fiber EMG electrode.

1 | INTRODUCTION

Single-fiber electromyography (SFEMG) uses a recording electrode with a recording surface area of 0.0005 mm^2 that allows identification of action potentials (APs) from individual muscle fibers.[1] Concentric needle electrodes (CNEs) are now increasingly used instead of single-fiber electrodes (SFEs), due to concerns for transmittable diseases and the high cost to purchase, sterilize, and maintain the SFE. CNEs have significantly larger recording surface areas, being 0.019 mm^2 in the smallest CNE commercially available, often referred as the "facial" CNE. In this report, the term "CNE" refers specifically to this electrode. By increasing the low-frequency filter setting to 1 kHz or more it is possible to record signals that are similar or identical in appearance to single-fiber APs, so-called apparent single-fiber action potentials (ASFAPs). An international collaboration recently published reference values (RVs) for jitter in orbicularis oculi, frontalis, and extensor digitorum muscles using this CNE.[1] These RVs were generally lower than those derived from SFEs,[1-3] in contrast to other studies reporting that the jitter values derived from CNEs and SFEs were similar.[4,5] A practical question for an individual electromyographer would be: What RVs should I adopt for the electrode I am using?

The concept of deriving reference limits from laboratory measurements that contain normal and abnormal values was suggested by Hoffman in 1963.[6] This concept was modified by Jabre and colleagues and presented as the E-Norm method.[7] Nandedkar et al used the same principles but employed a different algorithm to define the reference limits; this was called the E-Ref procedure.[8] E-Norm and E-Ref each showed good concordance between institutional normal limits and established RVs.[7,8] In this study, we applied E-Ref to CNE jitter values obtained at a single institution to derive the cutoff value and compared it with published RVs.

2 | METHODS

The study was approved by the institutional review board of the University of Alabama at Birmingham. All patient data were recorded at the electromyography laboratory of the University of Alabama at Birmingham. Retrospective chart review was performed by two investigators (I.L., M.K.) on all patients who had jitter studies performed between January 2018 and July 2019. ASFAP pairs were acquired using an amplitude trigger. During analysis, peak detection was used to identify ASFAPs and measure their interpotential interval (IPIs) (Sierra Summit; Cadwell Corp, Kennewick, WA). Filter settings were set at 1 to 10 kHz. A 30G × 25 mm CNE with a recording area of 0.02 mm^2 (Part No. 9013S0012; Dantec DCN Disposable Concentric Needle Electrode; Natus Neurology, Middleton, WI) was used for all jitter recordings. Recording was performed in the frontalis muscle during voluntary contraction. Only potentials with a short rise time, a well-defined peak, and a stable shape on consecutive discharges were accepted and analyzed. For each pair of APs analyzed, a minimum of 50 discharges and an ideal of more than 100 discharges were recorded.

Demographic information and indications for the test were extracted and described. The cutoff value for abnormal jitter was calculated from mean consecutive difference (MCD) values of all ASFAP pairs using the E-Ref procedure described by Nandedkar et al[8] and compared with the recently published CNE RV (38 microseconds).[1] After noting that the E-Ref–derived cutoff value differed from the published CNE RV, the authors (I.L., M.K.) reviewed all recorded ASFAP pair waveforms to determine whether the discrepancy was due to different criteria for ASFAP quality. Recordings that did not meet the published signal quality criteria regarding composite peaks, unstable amplitudes, riding signals, and false triggering were deleted.[1,9] Recordings with fewer than 50 discharges after deleting suboptimal recordings were also rejected. MCD values were calculated for recordings before and after the waveform review. The number of rejected MCD values, change in MCD values after review, and whether the change may have affected the conclusions of the test were determined. Cutoff values for abnormal jitter were then calculated from MCD values of acceptable ASFAP pairs using the E-Ref procedure, as described elsewhere.[8] In brief, the MCD values were normalized, organized in ascending order of their value, and then plotted against their normalized measurement rank. E-Ref is the value where the plot shows the greatest change in angle. The proportion of normal and abnormal jitter values determined by the E-Ref–derived cutoff values, published CNE RV (38 microseconds)[1] and published SFE RV (50 microseconds)[3] were compared using the chi-square test. The smallest published value for the frontalis muscle (50 microseconds) was used as the SFE upper limit of individual pairs.[3] Statistical analysis was done using R version 3.3.2 (R Foundation for Statistical Computing, Vienna, Austria). $P < .05$ was considered statistically significant.

3 | RESULTS

Results of 79 jitter studies from 77 patients were included. The mean age of the patients was 57 (standard deviation [SD], 16.1) years, and 53 (69%) were women. Most studies (74, or 96%, were requested as part of the diagnostic evaluation; 3 (4%) were performed to evaluate the severity of disease. The average number of potential pairs recorded in each study was 19 (SD 3.5).

A total of 1501 ASFAP pairs were included, with a median MCD of 21 (interquartile range [IQR], 15-30) microseconds. After review, 1371 (91.3%) ASFAP pairs were determined to have acceptable quality (median MCD, 19 [IQR, 15-28] microseconds) and 130 pairs were rejected (23 pairs with MCD value over 38 microseconds). Among those accepted, 220 pairs had MCD value changes of greater than 1 microsecond after review, and in 22 pairs the MCD value changed from greater than 38 microseconds to less than 38 microseconds (21 pairs) or vice versa (1 pair). Thirty studies included either rejected pairs with MCD value greater than 38 microseconds or pairs in which the MCD value changed to less than or greater than 38 microseconds. The proportion of pairs with MCD above 38 microseconds changed from over 10% to less than 10% in one study and vice versa in another study after the review, changing the conclusion of these studies.

The cutoff values derived by E-Ref from all ASFAP pairs before (Figure 1) and after signal review were 36 and 35 microseconds, respectively. The proportion of abnormal pairs identified by RVs from signals before or after review was not significantly different from the proportion of abnormal pairs determined by the published multicenter study RV for CNE (38 microseconds) (Table 1).[1]

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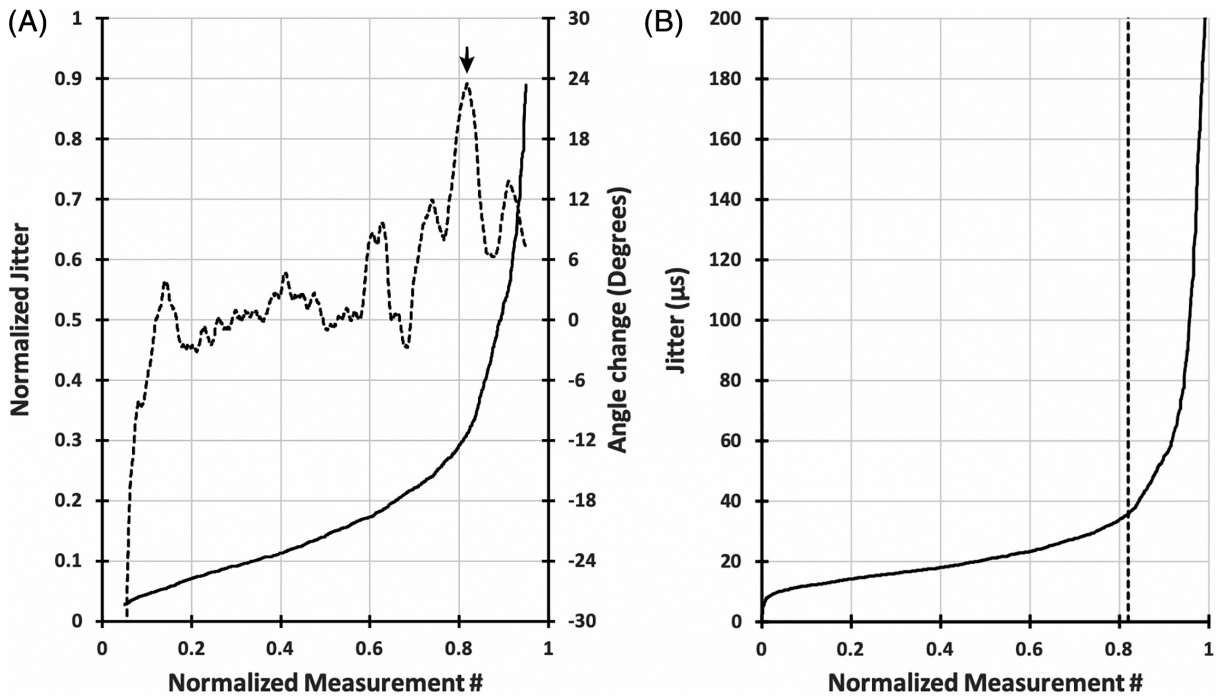

FIGURE 1 E-Ref procedure using jitter values before the review of waveforms. A, The normalized jitter values (y axis on the left) are plotted against normalized measurement (solid trace). The change in angle (y axis on right) shows a peak (arrow) that identifies E-Ref. B, Jitter values, sorted in ascending order, are plotted against the normalized measurement number. The vertical dashed cursor defines the upper normal limit using E-Ref. Abbreviation: E-Ref, extrapolated reference values procedure

4 | DISCUSSION

Electrodiagnostic laboratories often use reference values developed elsewhere despite potential differences in patient populations, EMG equipment, electrodes, and technique.[10] Although developing institutional reference values is preferred, this is often impractical due to the required resources or ethical issues. The E-Ref procedure was developed to aid when it is not possible or practical for individual laboratories to develop RVs,[11] and to provide a way for individual examiners to compare values obtained from their patient population vs published RVs.

The small difference between the E-Ref and published CNE RV values may have stemmed from differences in the patient population, the technique of the examiner, or the methodology of establishing the upper limit. Our RV derived from MCD values before the review is identical to the upper limit of the 18th MCD value from a single-center study that obtained reference values from a healthy population.[12]

Reference values directly affect the determination of normal vs abnormal jitter, which is critical in interpreting the study. We demonstrated that the proportion of recordings identified as having abnormal jitter is significantly reduced if an SFE-derived cutoff value is adopted, which would result in reduced sensitivity of the test.[13] This could be clinically misleading, as jitter studies are commonly believed to be very sensitive in detecting neuromuscular junction disorders, and the possibility of having such a disease could be rejected after a negative study. We did not perform a detailed chart review to confirm the diagnosis of myasthenia gravis in the patients in this study, 32% of whom were not clinically evaluated at our institution. Furthermore, the diagnosis of myasthenia gravis commonly depends on jitter results when serologic tests and repetitive nerve stimulation tests are negative. Hence, it may not be possible to ascertain the final diagnosis independent of jitter results in this referred population.

Despite the fact that both examiners were trained in jitter analysis during their neuromuscular medicine fellowships, stringent review of the signals that were initially accepted revealed that approximately 9% of the recorded pairs had suboptimal quality, affecting the results in a few studies. Jitter recording is a time-consuming and labor-

TABLE 1 Proportion of abnormal ASFAP pairs from frontalis muscle based on different cutoff values

	E-Ref values (35 or 36 µs)	CNE RV (38 µs)	SFE RV (50 µs)	*P* value[a]	*P* value[b]
All ASFAP pairs (1501)	265 (18%)	244 (16%)	175 (12%)	.3	<.0001
Qualified ASFAP pairs (1371)	213 (16%)	191 (14%)	143 (10%)	.3	<.0001

Abbreviations: ASFAP, apparent single-fiber action potential; CNE, concentric needle electrode; E-Ref, extrapolated reference values procedure; RV, reported reference values; SFE, single-fiber EMG electrode.
[a]E-Ref value vs CNE RV.
[b]E-Ref value vs SFE RV.

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intensive procedure that requires collaboration of both examiner and patient. The examiner should adhere to the acceptable ASFAP criteria to avoid producing misleading test results. Individual laboratories need to ensure that the equipment and technique used strictly follow those of the study in which the RV was generated. Our study confirms that the E-Ref technique can be used to examine the compatibility of an RV with results from an individual laboratory.

CONFLICT OF INTEREST

I.L. has received funding from the Myasthenia Gravis Foundation of America, not related with this project. D.B.S. is a consultant to Accordant Health Service, Momenta Pharmaceuticals, Kashiv Biosciences, and Janssen. S.D.N., an employee of Natus Neuro, he has no conflicts of interest to report. M.K has no relevant disclosures.

ETHICAL PUBLICATION STATEMENT

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

REFERENCES

1. Stålberg E, Sanders DB, Ali S, et al. Reference values for jitter recorded by concentric needle electrodes in healthy controls: a multicenter study. *Muscle Nerve*. 2016;53:351-362.
2. Ad Hoc Committee of the AAEM Special Interest Group on Single Fiber EMG. Single fiber EMG reference values: a collaborative effort. *Muscle Nerve*. 1992;15:151-161.
3. Bromberg MB, Scott DM, Single fiber EMG reference values: reformatted in tabular form. AD HOC Committee of the AAEM Single Fiber Special Interest Group. *Muscle Nerve*. 1994;17:820-821.
4. Ertas M, Baslo MB, Yildiz N, Yazici J, Oge AE. Concentric needle electrode for neuromuscular jitter analysis. *Muscle Nerve*. 2000;23:715-719.
5. Sarrigiannis PG, Kennett RP, Read S, Farrugia ME. Single-fiber EMG with a concentric needle electrode: validation in myasthenia gravis. *Muscle Nerve*. 2006;33:61-65.
6. Hoffmann RG. Statistics in the practice of medicine. *JAMA*. 1963;185:864-873.
7. Jabre JF, Pitt MC, Deeb J, Chui KK. E-norms: a method to extrapolate reference values from a laboratory population. *J Clin Neurophysiol*. 2015;32:265-270.
8. Nandedkar SD, Sanders DB, Hobson-Webb LD, Billakota S, Barkhaus PE, Stålberg EV. The extrapolated reference values procedure: theory, algorithm, and results in patients and control subjects. *Muscle Nerve*. 2018;57:90-95.
9. Sanders DB, Arimura K, Cui L, et al. Guidelines for single fiber EMG. *Clin Neurophysiol*. 2019;130:1417-1439.
10. Dillingham T, Chen S, Andary M, et al. Establishing high-quality reference values for nerve conduction studies: a report from the normative data task force of the American Association of Neuromuscular & Electrodiagnostic Medicine. *Muscle Nerve*. 2016;54:366-370.
11. Pitt MC, Jabre JF. Determining jitter values in the very young by use of the e-norms methodology. *Muscle Nerve*. 2017;55:51-54.
12. Kouyoumdjian JA, Stålberg EV. Concentric needle jitter on voluntary activated frontalis in 20 healthy subjects. *Muscle Nerve*. 2013;47:440-442.
13. Benatar M, Hammad M, Doss-Riney H. Concentric-needle single-fiber electromyography for the diagnosis of myasthenia gravis. *Muscle Nerve*. 2006;34:163-168.

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Reference jitter values for the sternocleidomastoid muscle with concentric needle electrodes

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Abstract

Background: The aim of this study was to establish reference jitter values for the voluntary activated sternocleidomastoid (SCM) muscle using a concentric needle electrode (CNE).

Abbreviations: ASFAP, apparent single fiber action potentials; BoNT, botulinum toxin; CNE, concentric needle electrode; MCD, mean consecutive difference; MIPI, mean interpotential interval; MG, myasthenia gravis; SFE, single fiber electrode; SFEMG, single fiber electromyography; SCM, sternocleidomastoid.

A part of the study has been presented at 32th National Clinical Neurophysiology EEG-EMG Congress, Mugla, Turkey, as an oral presentation.

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